

# On selective LF wh-movement: What Superiority and wh-islands tell us about wh-in-situ in Japanese

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**Abstract:** The paper examines wh-in-situ in Japanese and argues for a position that is in between wholesale LF wh-movement approaches, where all in-situ wh-phrases in Japanese undergo movement in LF, and the treatments where there is no LF wh-movement in Japanese at all by arguing that some, but not all wh-in-situ questions in Japanese involve LF wh-movement. Regarding when Japanese has LF wh-movement, Takahashi (1993) argues that in some cases, wh-scrambling in Japanese involves true overt wh-movement. I show that a rather intricate paradigm regarding wh-island and superiority effects with Japanese wh-in-situ can be accounted for if Japanese has LF wh-movement only in the contexts where Takahashi (1993) claims Japanese has overt wh-movement. The paper also establishes a parallel with wh-questions in French, providing a uniform treatment of Japanese and French, where (LF) wh-movement, Agree, and unselective binding are all involved in question formation, with Agree being clause-bounded, and unselective binding, a semantic operation, being available only after the +wh-C establishes a syntactic relationship with a wh-phrase, through either Agree or wh-movement. The discussion in the paper, which establishes the existence of LF superiority effects, also has consequences for our understanding of superiority and wh-island effects.

**Keywords:** Agree, LF wh-movement, superiority effects, unselective binding, wh-in-situ, wh-islands

## 1. Introduction

How to analyze wh-in-situ has been a hotly debated question. This paper examines wh-in-situ in Japanese. Taking as the point of departure analyses where all wh-phrases in-situ in Japanese undergo LF wh-movement, and those where none of them do, the paper takes an intermediate position: some, but not all, wh-in-situ questions in Japanese involve LF wh-movement. The claim will be based on a parallelism with an independently made claim regarding where Japanese has overt wh-movement made in Takahashi (1993) as well as a parallelism with wh-questions in French. The empirical basis of the paper will concern certain contrasts regarding wh-island and superiority effects with wh-in-situ in Japanese. It is, however, important to note that the scope of the paper will be limited to the data in question.<sup>1</sup>

The paper is organized as follows. Section 2 introduces the basic paradigm the paper will be dealing with, which concerns wh-island and superiority effects with wh-in-situ in Japanese. Section 3 examines how the data in question can be handled under various approaches to Superiority, under the assumption that

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\*For helpful comments, I thank the participants of my 2025 syntax seminar at the University of Connecticut, especially Masako Maeda.

The paper is dedicated to Myung-Kwan Park for everything he has done for Korean linguistics and the field more generally, and for 35 years of knowing each other, ever since my arrival at UConn in 1991.

<sup>1</sup>I will thus not attempt to address/evaluate arguments for different treatments of Japanese wh-in-situ given in the literature, which would take at a least a book-length work.

all Japanese questions involve LF wh-movement, demonstrating the problematic nature of the data under consideration under that analysis. Section 4 discusses Takahashi's (1993) analysis of when Japanese has overt wh-movement and section 5 makes a selective LF wh-movement proposal based on a certain parallelism with Takahashi's analysis. Section 6 establishes another parallelism, this time involving wh-questions in French, proposing a unified account for French and Japanese. Section 7 concludes the discussion. Finally, the appendix addresses a long-standing puzzle regarding interaction between superiority effects and interpretation of questions.

## 2. Wh-island and superiority effects with wh-in-situ in Japanese

Nishigauchi (1990) made an influential observation that Japanese shows wh-island effects based on examples like (1). Given that *-ka* can function either as a wh-question or a yes-no question complementizer, the readings in A-C should all be in principle available for (1).<sup>2</sup>

(1) [CP [IP Mai-wa [CP [IP dare-ga nani-o katta ka] sittei-mas-u ka].

Mai-TOP who-NOM what-ACC bought Q know-POL-PRS Q

A. OK: Does Mai know [who bought what]?

B. ??Intended: 'Who does Mai know [what (*t*) bought]?'

('For which *x*, *x* is a person, does Mai know what *x* bought?')

C. \*Intended: 'What does Mai know [who bought (*t*)]?'

('For which *y*, *y* is a thing, does Mai know who bought *y*?')

(Noguchi 2024, based on Nishigauchi 1990)

However, the only fully acceptable interpretation is A, where the two wh-phrases take scope in the most deeply embedded clause. The other interpretations, where one wh-phrase takes scope in the middle clause over the most deeply embedded question, are degraded. Since these interpretations involve interpreting a wh-phrase outside of a wh-island, Nishigauchi concludes that Japanese wh-in-situ questions show wh-island effects, which in turn provides evidence for LF wh-movement. However, there is more to this. Noguchi (2024) gives judgments that indicate that reading B is less degraded than reading C. Nishigauchi himself also observes this, and similar observations were made in Kurata (1991), Watanabe (1992), Richards (1997) (according to Watanabe 1992, Saito 1989b also observes the contrast in question). Two additional informants I have consulted, both linguists (Masako Maeda and Giulio Muramatsu), also find reading B to be better than reading C (although still degraded). In this paper I will focus on the contrast

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<sup>2</sup>For ease of exposition, for now I am ignoring the reading where both wh-phrases take the matrix scope, which is also degraded (the relevant reading will be discussed below).

What is also relevant is that the interpretation C improves if the object wh-phrase is scrambled over the subject wh-phrase, as in (2). Here, the interpretation C is judged to be on a par with the interpretation B from (1). In fact, the interpretation B is now more degraded (see also Saito 1994).<sup>3</sup>

“\*For which  $x$ ,  $x$  is a person, does Mai know what  $x$  bought?”

(4) ??Who does Mai know what bought

(6) \*Who does Mai know what bought

b. \*wh-Y > wh-X

3

The contrast in question is reminiscent of a superiority effect, illustrated by (8): given two wh-phrases in situ, if one of them undergoes wh-movement it must be the higher one—wh-moving a wh-phrase over a wh-phrase in an argument position results in degradation.

(8) ?\*What did who buy

Continuing to assume LF wh-movement for Japanese wh-in-situ, the contrast between a. and b. in (7) intuitively indicates the presence of a superiority effect. In fact, not contesting Nishigauchi's claim that in all the cases we are dealing with a wh-island effect, we can capture the full pattern if (7a) involves a wh-island violation, and (7b) both a Superiority and a wh-island violation.<sup>4</sup> Both (7a) and (7b) then involve the same violation, and (7b) involves another violation on top of that one, exactly the pattern where one degraded sentence is supposed to be worse than another one (as discussed in Epstein 1990).<sup>5</sup>

The suggestion is then that there is a wh-island violation in both (3)/(7b) and (4)/(7a) and additionally a Superiority violation in (3)/(7b) (still working with the underlying assumption that there is LF wh-movement in Japanese; this is regarding example (1), we will see that all of this extends to example (2), and the contrast between the interpretations in (5)-(6) regarding this example). While the intuition is clear the problem is that standard approaches to Superiority cannot give us this effect, as shown in the following section.

### 3. Different approaches to Superiority (assuming LF wh-movement)

Consider first the minimalist Attract Closest approach to Superiority (see Bošković 1997, 1998b, Cheng 1997, Kitahara 1993, and Oka 1993, among others), where, given two wh-phrases in situ, a wh-C attracts

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<sup>4</sup>Chomsky (1995) suggests an account that essentially unifies Superiority and wh-islands, but there is a clear empirical reason not to unify the two: the former results in a stronger violation than the latter, as (i) shows, (ia) being worse than (ib), hence I will assume below that we are dealing here with distinct conditions. (The discussion in this paper can in fact be interpreted as another argument that Superiority and wh-islands should not be unified).

(i) a. ?\*What did who buy

b. ??What do you wonder who bought

Regarding wh-islands, I will not be concerned with the exact technical implementation of the effect, simply assuming that movement out of an indirect question induces a wh-island effect. (Notice that the effect arises even when there is no wh-phrase in SpecCP—the presence of a wh-C is enough—as in ??*what do you wonder if/whether she saw* and (41)-(42) below). It is important to bear this in mind. (Regarding Superiority, see fn 6 and more generally section 3.)

<sup>5</sup>There is an argument frequently made by psycholinguists against generative syntax in general based on an alleged claim that it cannot account for degrees of ungrammaticality. This paper can be taken as a demonstration that the claim is completely unfounded, which is significant given the persistence of the claim (there are other works that can serve the same purpose, to mention just two, see Bošković 2023 and Chomsky 1986; for a more general discussion of degrees of ungrammaticality, see Epstein 1990).

the closest wh-phrase to it, hence superiority effects result when the wh-phrase that moves to wh-C is not the closest wh-phrase.<sup>6</sup> In light of this, consider the input for LF wh-movement in (3)/(7b) and (4)/(7a) (all these concern example (1)).

(9) [<sub>wh-CP</sub> [<sub>wh-CP</sub> [<sub>IP</sub> wh-X > wh-Y

In (4)/(7a), when the lower wh-phrase moves in LF across the higher wh-phrase to the embedded interrogative SpecCP we get a Superiority violation, on a par with (8), shown in step a in (10). In addition, when the wh-phrase not located in interrogative SpecCP in (10a) moves to the higher interrogative SpecCP in LF we get a wh-island violation (see fn 6 regarding Superiority), shown in step b in (10). (In other words, we get combined violations from (ia) and (ib) from fn 4).

(10) a. [<sub>wh-CP</sub> what<sub>j</sub> [<sub>IP</sub> who t<sub>j</sub> ]]

b. [<sub>wh-CP</sub> who<sub>i</sub> [<sub>IP</sub> ... [<sub>wh-CP</sub> what [<sub>IP</sub> t<sub>i</sub> ...

In (3)/(7b), on the other hand, the higher wh-phrase, who, moves to the lower interrogative SpecCP without violating Superiority, as shown in step a in (11). Subsequent movement of the lower wh-phrase, what, to the matrix SpecCP then only violates wh-islands, as shown in step b in (11). (So, we only get the violation (ib) from fn 4).

(11) a. [<sub>wh-CP</sub> who<sub>j</sub> [<sub>IP</sub> t<sub>j</sub> what ]]

b. [<sub>wh-CP</sub> what<sub>i</sub> [<sub>IP</sub> ... [<sub>wh-CP</sub> who<sub>j</sub> [<sub>IP</sub> t<sub>j</sub> t<sub>i</sub> ...

Under this analysis, then, (4)/(7a) violates both Superiority and wh-islands and (3)/(7b) violates only wh-islands, which is exactly the opposite from what we want (regarding example (1)). It is easy to verify that the same problem arises with example (2), whose relevant readings are given in (5)-(6), with reading (5) incorrectly predicted to be worse than reading (6) since the former involves a wh-island as well as a Superiority violation, while the latter involves only a wh-island violation.

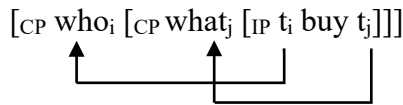
Chomsky's (1973) original approach to Superiority does not fare any better. The crossing/nested paths approach to Superiority (see e.g. Pesetsky 1982) also does not work. Under this approach, multiple wh-movement is fine if it results in nested paths but not if it results in crossing paths. Simplifying somewhat,

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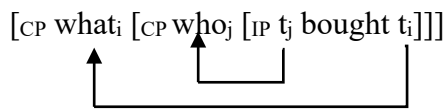
<sup>6</sup>Given that superiority effects and wh-islands should not be unified (see fn 4), wh-phrases in interrogative SpecCP are not candidates for wh-movement since wh-movement from interrogative SpecCP is not possible, hence they are ignored when it comes to superiority effects.

in (8) overt wh-movement of *what* results in the path shown in (12), and LF movement of *who* results in the path also shown in (12) (for ease of exposition, assume that LF wh-movement involves adjunction to CP). In acceptable (13), where *who* undergoes overt wh-movement and *what* undergoes LF movement, we have the paths shown in (13). (12) involves crossing paths and (13) involves nested paths. Since nested, but not crossing paths are allowed (12) is ruled out and (13) is not.

(12) \*What did who buy

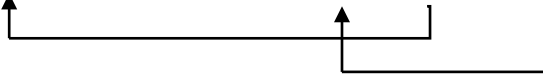


(13) Who bought what



This approach would also wrongly predict that (4)/(7a) should be better than (3)/(7b) since the derivation of the latter would violate Superiority and the derivation of the former would not (both would be wh-island violations) since the two would result in the paths shown in (14) and (15) respectively. As the reader can verify, the same problem again also arises with (5)-(6), the relevant readings for example (2).

(14) ??[<sub>CP</sub> who<sub>i</sub> does Mai know [<sub>CP</sub> what<sub>j</sub> [<sub>IP</sub> t<sub>i</sub> bought t<sub>j</sub>]]]



(15) \*[<sub>CP</sub> what<sub>i</sub> does Mai know [<sub>CP</sub> who<sub>j</sub> [<sub>IP</sub> t<sub>i</sub> bought t<sub>j</sub>]]]



There are also word order preservation approaches to Superiority where the idea is that wh-movement needs to preserve the word order of wh-phrases prior to wh-movement (see e.g. Müller 2000, 2001). In (13), it does (under the assumption that *who* moves to SpecCP), in (8) it does not, since *who* precedes *what* prior to wh-movement, but *what* precedes *who* after wh-movement. Such approaches are irrelevant to what we are discussing regarding Japanese wh-questions since all the actions there is taking place in LF, there are no word order changes of any sort. So, while the intuition discussed above regarding (4)/(5)/(7a) and (3)/(6)/(7b) is rather clear, the problem is that it cannot be implemented under any approach to Superiority.

The issue is that in the discussion above I was assuming that all questions in Japanese involve LF wh-movement, a traditional analysis that goes back to Lasnik and Saito (1984, 1992) and Nishiguachi 1990, and which was recently revived in Saito (2017) and Oda (2022). There are also approaches where Japanese

questions do not involve any kind of movement, like Shimoyama (2006). It is very difficult to see how the above data can be captured under such an analysis. I take Superiority and *wh*-islands to be movement violations, which means that we need movement involved in the constructions under consideration. What I would like to argue for is a position in between the wholesale LF *wh*-movement and no-*wh*-movement approaches, where some, but not all *wh*-in-situ questions involve LF movement. Such a position was actually argued for in Noguchi (2024), though based on very different considerations and on very different grounds from what will be proposed below.

Given that I will claim that only some Japanese *wh*-questions involve *wh*-movement, the immediate question that arises is in which context(s) Japanese has LF *wh*-movement. I will answer that question by setting up two parallelisms, one will involve a parallelism with a claim made in Takahashi (1993) regarding when Japanese has overt *wh*-movement, and the other will involve a parallelism with French questions.

#### 4. Takahashi (1993) on overt *wh*-movement in Japanese

It is well-known that Japanese *wh*-phrases can undergo scrambling, which under the traditional approach to scrambling means that they can undergo overt movement. Takahashi (1993) provides strong evidence that some cases of *wh*-scrambling actually involve true *wh*-movement, of the kind found in English. In other words, he argues that Japanese has overt *wh*-movement, but only in some contexts. What I will show is that if Japanese has LF *wh*-movement only in the contexts where Takahashi (1993) claims Japanese has overt *wh*-movement, we can straightforwardly capture all the data patterns discussed above (as well as a number of additional data points introduced below).

Takahashi (1993) argues that in certain contexts Japanese has true *wh*-movement, establishing the following generalization on overt *wh*-movement in Japanese (the following slightly simplifies Takahashi's generalization to more directly reflect its empirical coverage for ease of exposition).

(16) Long-distance movement of a *wh*-phrase to the initial position of a clause headed by a [+WH] COMP counts as *wh*-movement in Japanese.

Given the importance of Takahashi's generalization for the analysis developed below I will briefly summarize some of Takahashi's arguments for (16). First, to illustrate the scope of Takahashi's generalization, consider (17). (17b) illustrates the well-known fact that *wh*-phrases can undergo scrambling in Japanese. Takahashi argues that (17d) differs from (17b) in that the movement of the phrase in (17d) is not scrambling but true *wh*-movement, in accordance with generalization in (16). (17d) involves long-distance movement to the initial position of a +*wh* clause, *-ka* being a +*wh*-C. (All the Japanese data discussed in this section except (17c-d) are taken from Takahashi 1993.)

- (17) a. John-ga [<sub>CP</sub> Mary-ga nani-o katta ka] sitteiru.  
 John-NOM Mary-NOM what-ACC bought Q knows  
 'John knows what Mary bought.'
- b. Nani<sub>i</sub>-o John-ga [<sub>CP</sub> Mary-ga t<sub>i</sub> katta ka] sitteiru.  
 'John knows what Mary bought'
- c. John-ga [<sub>CP</sub> Mary-ga nani-o katta to] sitteimasu ka.  
 John-NOM Mary-NOM what-ACC bought that knows.POL Q  
 'What does John know that Mary bought.'
- d. Nani<sub>i</sub>-o John-wa [<sub>CP</sub> Mary-ga t<sub>i</sub> katta to] sitteimasu ka  
 'What does John know that Mary bought?'

Turning to arguments Takahashi gives for (16), consider (18).<sup>7</sup>

- (18) a. John-wa [<sub>CP</sub> Mary-ga nani-o tabeta ka] siritagatteiru no?  
 John-TOP Mary-NOM what-ACC ate Q wants.to.know Q  
 'Does J. want to know what M. ate?/What does J. want to know whether M. ate?'
- b. Nani<sub>i</sub>-o John-wa [<sub>CP</sub> Mary-ga t<sub>i</sub> tabeta ka] siritagatteiru no?  
 'What does John want to know whether Mary ate?'

While (18a) is ambiguous in that the wh-phrase can be interpreted either in the matrix or the embedded SpecCP, in (18b) the embedded clause interpretation is impossible. (18b) also differs from (17b) in this respect. Takahashi observes that this can be straightforwardly accounted for given the well-established fact that a wh-phrase that moves overtly to interrogative SpecCP must be interpreted in LF in that SpecCP, if the wh-phrase in (18d) does not undergo scrambling but true wh-movement.

Takahashi provides another argument to this effect based on superiority effects, which are again illustrated by the data below: given two wh-phrases that are candidates for wh-movement (so not those located in interrogative SpecCP), it's the higher wh-phrase that undergoes wh-movement.

(19) Who<sub>i</sub> did you persuade t<sub>i</sub> to read what?

(20) ?\*What<sub>j</sub> did you persuade who to read t<sub>j</sub>?

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<sup>7</sup>Note that Takahashi (1993) was not concerned with the wh-island effect noted by Nishigauchi (1990), hence it is ignored in the summary of Takahashi's arguments below.



Consider now the following data. There are two *wh*-phrases in situ in (21). Moving the lower *wh*-phrase over the higher *wh*-phrase results in degradation (22a), which can be accounted for if the movement in question involves true *wh*-movement, on a par with (20). Notice that clear scrambling is possible here, as shown by (22b), which indicates that the *wh*-phrase in (22a) is not undergoing scrambling.

(21) John-ga dare-ni [Mary-ga nani-o tabeta to] itta no?

John-NOM who-DAT Mary-NOM what-ACC ate C said Q

‘Who did John tell that Mary ate what?’

(22) a. ??Nani-o John-ga dare-ni [Mary-ga  $t_i$  tabeta to] itta no?

Lit. What did John tell who that Mary ate?

b. Pizza-o John-ga dare-ni [Mary-ga  $t_i$  tabeta to] itta no?

Pizza-ACC John-NOM who-DAT Mary-NOM ate C said Q

Lit. Pizza, did John tell who that Mary ate?

Interestingly, no superiority effect arises in (23):

(23) Nani-o John-ga [dare-ga  $t_i$  tabeta to] itta no?

what-acc John-NOM who-NOM ate C said Q

Lit. What did John say that who ate?

The lack of a superiority effect in (23) can be accounted for on a par with Bošković’s (1997) account of the lack of a superiority effect in Spanish (24). The subject there stays in SpecvP (Bonet 1989, Uribe-Echevarria 1992, Suñer 1994, among many others). Bošković argues that the object first undergoes A-movement, in particular object shift, as a result of which it is higher than the subject prior to *wh*-movement, hence no Superiority violation occurs in (24).

(24) ¿Quién compró qué?

what bought who

‘Who bought what?’

Given the well-known fact that clause internal scrambling in Japanese can involve A-movement, Takahashi argues that in (23), the object first undergoes A-scrambling within its clause (see (25)), as a result of which

it is higher than the subject prior to undergoing wh-movement into the matrix clause. Hence no superiority effect arises in (23), in contrast to (22a) (and on a par with (24)).

- (25) Nani-o John-ga [t<sub>j</sub> dare-ga t<sub>i</sub> tabeta to] itta no?  
 what-acc John-NOM who-NOM ate C said Q

### 5. Selective LF wh-movement: parallel with Takahashi (1993)

The gist of Takahashi's analysis that is important for our purposes can be summarized as shown in (26): essentially, if a wh-phrase moves across a CP boundary into an interrogative clause, the movement in question involves overt wh-movement, not scrambling. In other words, the context in question enforces wh-movement.

- (26) a. [CP-Q wh]                      overt scrambling here is not overt wh-movement  
 b. [CP-Q wh<sub>i</sub> [CP t<sub>i</sub>]]              overt scrambling here is overt wh-movement

I propose a parallel analysis for wh-in-situ and LF wh-movement. Exactly in the context where overt movement of a wh-phrase counts as overt wh-movement in Japanese, Japanese must have LF wh-movement. Where overt movement of a wh-phrase does not count as wh-movement, Japanese does not have LF wh-movement. This is summarized below (for the moment we can assume that wh-licensing that does not involve LF wh-movement in Japanese involves unselective binding, but I will return to this issue below):

- (27) [CP-Q[CP wh]]              wh-in-situ here must involve LF wh-movement  
 (28) [CP-Q wh]                  no LF wh-movement here

The parallel between (26) and (27)-(28) should be obvious. The parallel also provides us with an immediate account of the data we have started the paper with. Recall that there is the following contrast for example (1) for the relevant interpretations ((29)-(30) indicate the relevant interpretations), with interpretation (29) being ?? and interpretation (30) \* for example (2).

- (29) \*What does Mai know who bought  
 (30) ??Who does Mai know what bought

The abstract pattern is repeated below: when wh-phrase X is higher than wh-phrase Y, interpreting one wh-phrase out of a wh-island is better if X is interpreted higher than Y.

(31) Overt Syntax positions: wh-X > wh-Y (> indicates height)

LF interpretation positions:

a. ??wh-X > wh-Y

b. \*wh-Y > wh-X

I have shown in section 3 that although intuitively, the pattern looks like it involves a superiority effect, it cannot be accounted for under any approach to Superiority discussed there. We will now see that the problem with the account considered in section 2 was the assumption that there is wholesale LF wh-movement in Japanese, which results in the structures in (32) (the original positions are in strikethrough, the LF positions without it):

(32) a. ??[<sub>Wh-CP</sub> wh-X [<sub>IP</sub> [<sub>Wh-CP</sub> wh-Y [<sub>IP</sub> ~~wh-X~~ ~~wh-Y~~]]]]

b. \*[<sub>Wh-CP</sub> wh-Y [<sub>IP</sub> [<sub>Wh-CP</sub> wh-X [<sub>IP</sub> ~~wh-X~~ ~~wh-Y~~]]]]

If LF wh-movement in Japanese takes place only in the context where overt movement of a wh-phrase in Japanese counts as wh-movement in Takahashi's analysis, the problem disappears.

Consider the representation of (29)-(30) before any LF wh-movement for example (1).

(33) [<sub>Wh-CP1</sub> Q [<sub>IP</sub> Mai knows [<sub>Wh-CP2</sub> Q [<sub>IP</sub> who bought what]]]]

The problem before was that the interpretation where *what* is interpreted in Wh-CP2 and *who* is interpreted in Wh-CP1 involved LF wh-movement of *what* over *who*, a Superiority violation (see step (10a)). Later LF wh-movement of *who* to Wh-CP1 added a wh-island violation (see step (10b)), as a result of which (30)/(31a) violated both Superiority and wh-islands. However, under the parallel with Takahashi's contexts, there is no LF movement of *what* to the most embedded SpecCP since we are dealing here with a single-clause environment, i.e. there is no step (10a). The only LF wh-movement involves *who*: *who* wh-moves in LF to the matrix SpecCP—since the movement takes place out of an indirect question, it violates wh-islands (see fn 4).

Regarding (29)/(31b), before it involved only a wh-island effect: there was no superiority effect since LF wh-movement of *what* did not cross another candidate for wh-movement, i.e. it did not cross a wh-phrase that is not located in SpecCP since *who* moves to Wh-CP2 before *what* moved to Wh-CP1 (see (11)). Now, there is no LF wh-movement of *who* to Wh-CP2. This is a context where no LF wh-movement is involved, as discussed above. Consequently, when *what* moves into the matrix clause, Wh-CP1, it crosses

another wh-phrase in situ, which results in a Superiority violation. Additionally, since this involves wh-movement out of an indirect question, we get a wh-island violation. We then get exactly what we need: (29)/(31b) violate both Superiority and wh-islands, while (30)/(31a) violate only wh-islands.

Consider now the structure for example (2) before any LF wh-movement takes place. The judgment pattern for different interpretations of example (2) is repeated in (35)-(36).

(34) [<sub>Wh-CP1</sub> Q [<sub>IP</sub> Mai knows [<sub>Wh-CP2</sub> Q [<sub>IP</sub> what<sub>i</sub> who bought t<sub>i</sub>]]]]

(35) ??What does Mai know who bought

(36) \*Who does Mai know what bought

We have seen above that the wholesale LF wh-movement gives a wrong result for (2), incorrectly predicting reading (35) to be worse than (36) (since under that account, (35) involves a wh-island and a Superiority violation and (36) only a wh-island violation). The problem does not arise under the selective LF wh-movement analysis proposed above, where there is no LF wh-movement to the embedded SpecCP in (2)/(34). Under that analysis, on the reading in (35), movement of *what* from the indirect question results in a wh-island violation—this is the only violation since Superiority is satisfied (*what* undergoes LF wh-movement from a position higher than *who*, see (34)). In (36), on the other hand, both Superiority and Wh-Islands are violated since movement of *who* over *what* results in a Superiority violation (given the pre-wh-movement structure in (34)), in addition to the wh-island violation that movement out of an indirect question induces.

Under the LF parallelism with Takahashi's account analysis, we thus explain not only why reading B is worse than reading C in (1), but also why the pattern is reversed in (2), where reading B improves and where reading C is worse than reading B, while also accounting for the fact that none of these readings is fully acceptable (see also fn 5 for a broader significance of this achievement).

Recall now that the superiority effect with overt wh-movement in Takahashi's Japanese data arose only in the context where the wh-phrases were in different clauses, cf. the contrast between (22a) and (23), which is not the case with the analysis presented above, where the superiority effect arises even in the context that parallels (23). Recall, however, that a step of A-scrambling, which precedes wh-movement, provided an escape hatch from Superiority in (23) (see (25)). This is obviously not an option for LF wh-movement, since it would require the scrambling movement to apply in LF.<sup>8</sup> We thus account for the full

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<sup>8</sup>Scrambling is standardly assumed to be an overt syntax operation. E.g., if scrambling were possible in LF, (ib) would be incorrectly predicted to be ambiguous, just like (ia), as a result of LF scrambling of the object.

(i) a. Daremo-ni dareka-ga t<sub>i</sub> atta.  
 everyone-DAT someone-NOM met  $\forall > \exists; \exists > \forall$

range of data discussed above, crucially relying on a parallelism between contexts where Japanese has overt and LF wh-movement: only in the contexts where movement of a wh-phrase counts as overt wh-movement does Japanese have LF wh-movement (see below for an account of this state of affairs).<sup>9</sup>

The existence of LF superiority effects, confirmed by the above discussion, is also of some theoretical significance. In particular, it argues against word order preservation approaches to Superiority since LF superiority effects do not affect word order.

## 6. Parallel with French

There is another parallel that is worth noting here. It is well-known that in principle, French allows both wh-movement and wh-in-situ. However, as Bošković (1998a) observes, this is the case only with single-clause questions. In long-distance questions, wh-in-situ is not possible, as shown by the data below.<sup>10</sup>

- (37) a. Tu as vu qui?  
           you have seen who  
           ‘Who did you see?’  
       b. Qui as-tu vu?
- (38) a. \*Jean et Pierre croient que Marie a vu qui?  
           Jean and Pierre believe that Marie has seen whom

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‘Everyone, someone met.’

- b. Dareka-ga daremo-ni atta.  
    someone-NOM everyone-DAT met      $\exists > \forall; * \forall > \exists$   
    ‘Someone met everyone.’

<sup>9</sup>Note that I do not adopt Takahashi’s (1994) claim, based on sluicing, that Japanese always has optional overt wh-movement, in fact, the discussion in this paper argues against it (see also Abe 2015, 2016 for a dissenting view on sluicing; see also the discussion in section 6).

It should also be noted that Richards (1997) offers an account of the contrast between the B and the C reading of (1) in his system, where IP is essentially an interrogative projection in Japanese though movement to SpecCP still takes place. The gist of the account is that on the best derivation for reading C, both Shortest Attract (the C fails to attract the closest wh-phrase) and Shortest Move (the wh-phrase skips a potential legitimate landing site) are violated, while on reading B only Shortest Move is violated, although there appear to be two such violations since both wh-phrases seem to violate it so it is not completely clear how this works (Richards consider Shortest Attract and Shortest Move the same condition (Shortest), effective on different elements/relations). Also, the system Richards adopts is quite different from the current one theoretically—he does not assume Criterial Freezing (Rizzi 2006) since he allows wh-movement from an interrogative SpecCP, though a version of Criterial Freezing holds, in that a wh-phrase that moves to SpecIP in Japanese cannot move from the interrogative SpecCP after subsequent movement from SpecIP to the clause-mate interrogative SpecCP. In a sense, then, for Richards Criterial Freezing holds for movement to SpecIP (since then it is not possible to scope out of the immediately dominating CP) but not for movement to SpecCP (you can scope out of that SpecCP, as long as the movement of the wh-phrase to SpecCP was not preceded by its movement to SpecIP). Note that the current work adopts Criterial Freezing in Rizzi’s sense (see below). For an empirical problem for Richards’s (1997) account, which does not arise under the current analysis, see footnote 17.

<sup>10</sup>See also Boeckx (2000), Bošković (2000), Cheng and Rooryck (2000), Mathieu (1999).

‘Who do Jean and Pierre believe that Marie saw?’

b. Qui Jean et Pierre croient-ils que Marie a vu?

(Bošković 1998a)

There is again an interesting parallel here:

-In French in overt syntax, wh-in-situ in long-distance questions is not possible; you must have overt wh-movement

-In Japanese in LF, wh-in-situ in long-distance questions is not possible, you must have LF wh-movement  
Suppose that true wh-in-situ, with no LF wh-movement, is licensed by unselective binding. One may then think that unselective binding is not possible across a clausal boundary here. But this cannot be right. (38a) contrasts with (39), where wh-in-situ within a long-distance question is possible. The difference between (38a) and (39) is that there is a higher wh-phrase in (39) which is a clause-mate with the +wh-C, which is not the case with (38a).

(39) Qui croit que Marie a vu qui?

who believes that Marie has seen who

(Bošković 1998a)

Another relevant example is given in (40), which patterns with (39) (the judgment for (40) is due to Éric Mathieu).

(40) Tu as dit à qui que Mary a vu qui?

you have told to who that Mary has seen who

‘Who did you tell that Mary saw?’

Japanese in fact behaves like French in the relevant respect. Consider Japanese *ka dooka* questions. *Ka dooka* is unambiguously a yes-no question complementizer, essentially a counterpart of English *whether*. This means that the wh-phrase in (41) can only be interpreted in the matrix SpecCP, a context which in Japanese requires LF wh-movement, as discussed above, hence the wh-island effect. Not surprisingly, the effect is also present in (42). Interestingly, it is not present in (43). The difference between (42) and (43) is that in (43), there is another wh-phrase that is a clause-mate with the +wh-C.

(41) ??John-wa [Mary-ga nani-o katta ka dooka] siritagatte iru no?

John-TOP Mary-NOM what-ACC bought whether know-want Q

‘What does John want to know whether Mary bought?’

(42) ??John-wa [Mary-ga nani-o katta ka dooka] Tom-ni tazuneta no?

John-TOP Mary-NOM what-ACC bought whether Tom-DAT asked Q

‘What did John ask Tom whether Mary bought *t*?’

(43) John-wa [Mary-ga nani-o katta ka dooka] dare-ni tazuneta no?

John-TOP Mary-NOM what-ACC bought whether who-DAT asked Q

‘Who did John ask *t* whether Mary bought what?’ (Watanabe 1992)

The contrast between (42) and (43) parallels the contrast between French (38) and (39). It indicates that a *wh*-phrase is not forced to undergo LF *wh*-movement out of a CP if there is another *wh*-phrase that is a clause-mate with the +*Wh*-C.<sup>11</sup>

That not all *wh*-phrases are forced to move in LF out of their CP is confirmed by (44). (44) has the same status as (41)-(42). This is a context where LF *wh*-movement is forced, as discussed above. If both *wh*-phrases had to undergo LF *wh*-movement we might expect (44) to be worse than (41)-(42) due to the cumulative effect of locality violations (discussed above). If only one *wh*-phrase has to undergo LF-movement here, then it is not surprising that (44) is on a par with (41)-(42).

(44) ??John-wa [dare-ga nani-o katta ka dooka] Tom-ni tazuneta no?

John-TOP who-NOM what-ACC bought whether Tom-DAT asked Q

‘What did John ask Tom whether who bought *t*?’ (Watanabe 1992)

How can we make sense of these data patterns? Consider first French. Let us assume that the +*wh*-C in French must establish a syntactic dependency with a *wh*-phrase, but that this can be done either through Agree or *wh*-movement, i.e. the traditional Spec-head relationship (SHA).<sup>12</sup> I furthermore assume that Agree is clause-bounded, possibly as a result of a PIC effect.<sup>13</sup> In a single-clause environment, both Agree and *wh*-movement/SHA are options, the latter taking place in the *wh*-movement construction in (37b) and the former in the *wh*-in-situ construction in (37a). As for long-distance questions, Agree is not an option

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<sup>11</sup>See Richards (1997) for another perspective on these data. Note that there is some disagreement regarding the exact status of (43) if *dare-ni* precedes the *ka dooka* clause, compare Watanabe (1992) with Maki (1995) and Richards (1997) (see also Hagstrom 1998:92-92) so I will put such examples aside here.

<sup>12</sup>Bošković (2025) recasts SHA in terms of licensing under sisterhood, which he argues is needed independently of Agree. Anyway, the exact implementation of the movement option does not matter for us here.

<sup>13</sup>If this is a PIC effect, this would mean that vP is not a phase, as several authors have argued (e.g. Keine 2017, Keine and Zeijlstra 2025); see also Bošković (2007) for an opposing view on Agree and the PIC, where Agree is not subject to the PIC. As discussed in Bošković (2003), under this view Agree into a clause can be blocked via Agree Closest by the CP that dominates the relevant element. Under this view, whether or not vP is a phase would not matter. At any rate, the details do not matter for us, what matters simply is that Agree is clause-bounded, i.e. that Agree into a CP is not possible.

here, Agree being clause-bounded. As a result, only wh-movement is possible in this context. In other words, (38b) involves wh-movement/SHA while (38a) is ruled out due to a locality constraint on Agree. Why is (39) acceptable, in contrast to (38a)? It is because, in contrast to (38a), in (39) there is a higher wh-phrase with which wh-C can establish a syntactic dependency, in fact through either Agree or wh-movement/SHA ((40), on the other hand, involves Agree). Since the formal inadequacy of the wh-C that requires a syntactic dependency with a wh-phrase is satisfied, the wh-C and the wh-phrase in situ can undergo unselective binding, unselective binding being a purely semantic relationship that is not subject to syntactic locality considerations (it is not clause-bounded or subject to islandhood; note that *what* in English (45) and (46) is also licensed via unselective binding (given the presence of a higher wh-phrase), hence there is no clausematehood or islandhood effect).

(45) Who believes that Mary likes what?

(46) Who fell asleep after Mary had bought what?

(47) cf. ??What did you fall asleep after Mary had bought what?

The French data under consideration are then straightforwardly captured. The account in fact readily extends to the Japanese paradigm, the only modification being that in Japanese, wh-movement takes place in LF (I assume that Agree is an overt syntax operation,<sup>14</sup> hence there is no difference in its timing between French and Japanese). Thus, the wh-phrase within the *ka dooka* clause is forced to undergo LF wh-movement in (41)-(42) (resulting in a wh-island violation), but not in (43), since there, the matrix C can undergo Agree with the matrix clause wh-phrase (given that Agree is an overt syntax operation, LF wh-movement will take place only if Agree is not an option). The formal inadequacy of the matrix +wh-C being taken care of, it can then unselectively bind the wh-phrase within the *ka dooka* clause, which is not an option in (41)-(42).

As for (1), repeated below, assuming that wh-movement to an interrogative SpecCP freezes the wh-phrase in that position but that Agree with a wh-C does not have such a freezing effect (cf. in fact the Criterial Freezing effect of Rizzi 2006, which holds only for movement/SHA), the embedded wh-C can undergo Agree with the subject wh-phrase (which would not freeze the wh-phrase for LF wh-movement), or it could in fact undergo Agree with both the subject and the object wh-phrase, given that Multiple Agree is an option in Japanese (see Hiraiwa 2001).

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<sup>14</sup>In fact, Agree-licensed feature valuation has PF effects.



(1) [CP [IP Mai-wa [CP [IP dare-ga nani-o katta ka] sittei-mas-u ka.

Mai-TOP who-NOM what-ACC bought Q know-POL-PRS Q

A. OK: Does Mai know [who bought what]?

B. ??Intended: 'Who does Mai know [what (*t*) bought]?'

('For which *x*, *x* is a person, does Mai know what *x* bought?')

C. \*Intended: 'What does Mai know [who bought (*t*)]?'

('For which *y*, *y* is a thing, does Mai know who bought *y*?')

At any rate, nothing relevant changes in the derivations/interpretations discussed above for example (1). As noted above, given that Agree is an overt syntax operation, LF wh-movement will take place only if Agree is not an option. Since it is an option in a clause-mate context, this context will always involve Agree, in contrast to French, where Agree and wh-movement take place on the same cycle.<sup>15</sup> Long-distance context, on the other hand, will involve LF wh-movement, since Agree is not an option there.<sup>16</sup>

Regarding the multiple vs single Agree options noted above regarding (1) (the issue being whether the embedded wh-C undergoes Agree with *dare* or with both *dare* and *nani*), there is an example from Saito (1994) that may favor the multiple Agree option. Saito observes that (48) marginally allows reading (49a), but not (49b).<sup>17</sup>

(48) Kimi-wa [CP[IP[CP John-ga nani-o katta to]<sub>i</sub> dare-ga t<sub>i</sub> itta]] ka] siritai no

you-top John-nom what-acc bought Comp who-nom said Q want-to-know Q

'Q you want to know [Q [[CP that John bought what]<sub>i</sub> who said t<sub>i</sub>]]

(49) a. *Nani* takes matrix scope, and *dare* takes embedded scope

<sup>15</sup>Regarding reading C of (1), notice that if there were an option of *dare* 'who' undergoing LF wh-movement, when *nani* 'what' moves out of the embedded clause Superiority would not be violated, only the wh-island constraint would be. This means that LF wh-movement cannot be an option in a short-distance context.

Traditional LF movement is sometimes treated in terms of lower copy pronunciation, i.e. as overt movement+lower copy pronunciation. That would not work here because of the timing: the relevant movement would then take place on the same cycle as Agree (as in French). What is needed here is in fact the traditional LF movement.

<sup>16</sup>The analysis accommodates in a natural way Takahashi's (1993) data/generalization regarding overt wh-movement in Japanese, it in fact accommodates the spirit of Takahashi's account of this generalization. As discussed in Saito (1989a, 1992), Tada (1993), Bošković and Takahashi (1998), among many others, long-distance scrambling is subject to obligatory radical reconstruction (see these works for empirical evidence to this effect). If the wh-phrase in LD contexts like (17) were to undergo scrambling, it would have to undergo radical reconstruction into the embedded clause only to then again undergo LF movement into the matrix clause. Takahashi argues that the movement of the wh-phrase in (17) counts as wh-movement since fewer instances of movement are needed to satisfy the relevant requirements than if it involved scrambling.

<sup>17</sup> Note that these data cannot be accounted for under Richards' (1997) analysis, summarized in fn 9.

(49a) is not surprising, in fact it does not differ from (2), repeated below, in the relevant respect: embedded wh-C undergoes Agree with *dare-ga* in (48), with *nani-o* undergoing LF wh-movement into the matrix clause, which violates only the wh-island constraint.

(2) Mai-wa [nani-o<sub>i</sub> dare-ga t<sub>i</sub> katta ka] sittei-mas-u ka.

Mai-TOP what-ACC who-NOM bought Q know-POL-PRS Q

‘??For which y, y is a thing, does Mai know who bought y?’

‘\*For which x, x is a person, does Mai know what x bought?’

Now, in (2), as discussed above, the derivation on which *dare-ga* takes wide scope violates both Superiority (due to the presence of a higher wh-phrase, *nani-o*, which remains in situ), and wh-islands. In (48), on the other hand, since *nani-o* does not c-command *dare-ga* it appears that Superiority is not violated (for Superiority to be in effect, the crossed wh-phrase needs to c-command the moved wh-phrase prior to the movement, see e.g. Bošković 1997), which would leave us with only a wh-island violation here, an undesirable result. The Multiple Agree account suggested above for (1) can, however, account for the unacceptability status of the reading in question for (48) if we assume that if there is no wh-movement, only Agree, the wh-C must mark the scope of one wh-phrase it agreed with.<sup>18</sup> In (1), the embedded wh-C undergoes Multiple Agree with *dare-ga* and *nani-o*. (In (2), there is also Agree with both wh-phrases.) One of the wh-phrases then must be interpreted in the embedded clause, the other one is still free to be interpreted in the higher clause since Agree with a wh-C does not result in a freezing effect. The account, however, does not extend to (48)—the embedded wh-C cannot undergo Multiple Agree in (48): given that Agree is clause-bounded the embedded wh-C can only Agree with *dare-ga*, which is then forced to take scope in the most embedded clause. The reading in question is still somewhat degraded due to a wh-island effect.

The patterns discussed in this section may also have consequences for our understanding of the diachrony of question dependencies. The predecessors of both French (Latin) and Modern Japanese (Old Japanese) were multiple wh-fronting languages (see Ledgeway 2012, Dadan 2019 for Latin and Aldridge 2009, Dadan 2019 for Old Japanese; for Old Japanese see also Watanabe 2002, Whitman 2001). Both lost multiple wh-fronting. Japanese also lost overt wh-movement, except for the limited contexts discussed by Takahashi (1993), and French is clearly in the process of becoming a wh-in-situ language (see the corpus

<sup>18</sup>I mean here the wh-C, not yes-no question C.

studies in Ashby 1977, Coveney 1995). A number of interesting questions arise here, e.g. is LF wh-movement a step in the process of becoming fully a wh-in-situ language, with no syntactic dependency between interrogative +wh-C and wh-phrases (Mandarin Chinese, which also was a wh-movement language (Aldridge 2010, Dadan 2019), seems to have reached that stage; thus, there are no wh-island effects in Mandarin Chinese, see Huang 1982, Tsai 1994<sup>19</sup>). French, which appears to be undergoing an active diachronic change of losing overt wh-movement, may be particularly useful to investigate regarding the diachrony of question dependencies. If the optionality of overt wh-movement is an indication of its eventual loss, French may be indicating that wh-movement is first lost in matrix short-distance questions.<sup>20</sup> Given that diachronic changes involve children re-analyzing existing structures, what may matter here (at least when it comes to indirect questions, where wh-in-situ is still not possible) is the zero-degree-embedding hypothesis, which essentially confines (initially) language learning to basic, non-embedded sentences.

## 6. Conclusion

This paper has examined wh-in-situ in Japanese and argued for a position that is in between wholesale LF wh-movement approaches, where all in-situ wh-phrases in Japanese undergo movement in LF, and the treatments where there is no LF wh-movement in Japanese at all. In particular, I have argued that some, but not all wh-in-situ questions in Japanese involve LF wh-movement. Regarding when Japanese questions have LF wh-movement, I have argued for a parallel with Takahashi's (1993) claim regarding when Japanese has overt wh-movement. Takahashi (1993) argues that in some cases, wh-scrambling in Japanese actually involves true overt wh-movement, of the kind found in English. I have shown that a rather intricate paradigm regarding wh-islands and superiority effects with Japanese wh-in-situ can be accounted for in a principled way if Japanese has LF wh-movement only in the contexts where Takahashi (1993) claims Japanese has overt wh-movement, which is long-distance questions. In this respect, I have also argued for a parallel with wh-questions in French: In French in overt syntax, wh-in-situ in long-distance questions is not possible, you must have overt wh-movement. In Japanese in LF, wh-in-situ in long-distance questions is not possible, you must have LF wh-movement. I have provided a uniform treatment of French and Japanese, where (LF) wh-movement, Agree, and unselective binding are all involved, with Agree being clause-bounded, and unselective binding, which is a semantic operation, being available only after the +wh-C establishes a syntactic relationship with a wh-phrase, through either Agree or wh-movement. It should be noted that the discussion in this paper provides particularly strong evidence against approaches to Japanese

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<sup>19</sup>Though see Chen (2024) (for relevant discussion, see also Qilin, Park, and Yang 2022).

<sup>20</sup>Recall also that Japanese does not have even LF wh-movement in this context.

wh-in-situ where Japanese questions do not involve any kind of movement, like Shimoyama (2006). It is hard to see how the superiority/wh-island contrasts discussed in this paper can be captured under such an approach.<sup>21</sup> The analysis of wh-island/superiority effects with Japanese wh-in-situ presented in this paper can also be taken as a demonstration that generative syntax has tools to capture fine-grained degrees of ungrammaticality. (see footnote 5). Finally, the existence of LF superiority effects, revealed by the discussion in this paper, is of some theoretical significance—it argues against word order preservation approaches to Superiority since LF superiority effects do not affect word order. The discussion in this paper has also provided evidence that superiority and wh-island effects should not be unified.

### Appendix: A Superiority puzzle

Since a good chunk of this paper deals with superiority effects, in this appendix I will address a long-standing puzzle regarding superiority effects (for some discussion, see Lasnik and Saito 1992, Golan 1993, Fox 2000, Reinhart 2006, Bošković 2022). Consider the following data, noted by Lasnik and Saito (1992).

- (50) a. I wonder who bought what.  
b. \*I wonder what who bought.  
c. Who wonders who bought what?  
d. Who wonders what who bought?

(50a-b) display the familiar superiority effect. Interestingly, (50d) is grammatical in spite of *what* undergoing wh-movement across *who*. Importantly, it is not ambiguous: it is acceptable only if the second *who* takes matrix scope. So there is a superiority effect here, but it is only reflected in the interpretation. What these data seem to indicate is that Superiority is conditioned on the wh-phrases being interpreted in the same interrogative clause: (50d) is acceptable since on the acceptable reading the wh-phrases are not interpreted in the same interrogative clause.

This, however, raises a serious problem. The superiority effect is a result of a formal derivational requirement of the computational system. How can we bring in semantic considerations into a requirement of the computational system? What makes the situation even more puzzling is that at the point of movement of *what* to the embedded SpecCP, which takes place before the matrix clause is built in (50d), (50b) and (50d) are identical syntactically. Movement to the embedded SpecCP somehow needs to be made sensitive

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<sup>21</sup>In particular, the problem arises with respect to superiority effects (since the analysis can capture wh-island effects, see Shimoyama 2006). The same actually holds for Hagstrom's (1998) Q-particle movement analysis, on which the Q-particle rather than wh-phrases undergoes movement in Japanese.

to the final semantics of the examples in question. In addressing these data, Golan (1993), Fox (2000), Reinhart (2006) suggest that only constructions with essentially the same semantic interpretation can be compared regarding economy of derivation (i.e. Superiority), which really only restates the problem to be accounted for (in addition to involving a very significant look-ahead). In this appendix I will offer a brief speculation regarding these data.

Given that *wh*-phrases in situ in English are interpreted through unselective binding, as discussed above, intuitively it appears that unselective binding may matter here. I would like to suggest making a parallel with the mechanism of rescue by PF deletion. Ross (1969) observed that locality violations disappear if the island from which movement takes place is elided. This is illustrated by (51). (51a) involves movement out of a complex NP, the violation apparently disappears if the complex NP island is elided under sluicing, as in (51b).

(51) a. \*She kissed a man who bit one of my friends, but Tom does not realize which one of my friends she kissed a man who bit. (Ross 1969)

b. She kissed a man who bit one of my friends, but Tom does not realize which one of my friends.

Chomsky (1972) proposes an account of the effect the gist of which was subsequently adopted in a number of works (e.g. Merchant 2001, Lasnik 2001, Hornstein, Lasnik, and Uriagereka 2003, Bošković 2011): when movement out of an island takes place, the island is *\*-marked*. If the *\** is present in the final PF representation, we have a violation. If the *\*-marked* island is removed by PF deletion/ellipsis, no *\** is present in the final PF violation, hence there is no violation.

A similar account can be applied to the puzzling Superiority data in (50), the effect that we are trying to capture informally being that if I unselectively bind you, only then I know that you were a candidate for *wh*-movement. The rescue-by-PF-deletion account suggests that there is a representational aspect to locality violations, which is what I will appeal to in the account of (50). Superiority violations are locality violations, so let's apply the *\*-marking* mechanism to them, which was in fact proposed in Bošković (2011). Furthermore, suppose that what is *\*-marked* with Superiority violations is the *+wh-C* (contra Bošković 2011, where *wh*-phrases are *\*-marked*), which seems natural under the Attract closest account of Superiority since it is the *C* that fails to do something in the most economical way when Superiority is violated. Consequently, after *wh*-movement of *what* to the embedded SpecCP in (50d), the embedded *C* is *\*-marked*. The idea is then that if the clause headed by the relevant *C* is not interpreted as a multiple question, the *\** is ignored/deleted since it is only relevant to multiple question interpretation, i.e. when more than one *wh*-phrase is interpreted with the same *C*.

There is another way of implementing this. Suppose we assume necessity of Higginbotham and May's (1981) absorption for proper interpretation of multiple questions. The suggestion would then be that the presence of a \* blocks absorption, which means that a \*-marked question cannot be interpreted as a multiple question.<sup>22</sup> This blocks the possibility of *who* being interpreted in the embedded CP in (50d): movement of *what* across *who* results in \*-marking, which then blocks the impossibility of multiple question interpretation for this clause (by blocking quantifier absorption that is necessary for proper interpretation of multiple questions).

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<sup>22</sup>Lasnik and Saito (1992) also appeal to absorption but with a very different implementation.

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